Detailed 10-Week Execution Roadmap

Phase 1 — Development & Reproducibility (Weeks 1–4)

Week 1: Source Acquisition & Repository Setup

- **Tasks:** Acquire and document NCERT/PYQ sources. Clean text data and apply language tags. Stand up Git and Data Version Control (DVC) prior to touching data to ensure versioning is not retrofitted.
- **Deliverables:** Documented NCERT/PYQ Source Catalog, Cleaned & Language-Tagged Corpus, Operational Git + DVC Repository Architecture.
- **Roles:** Data Engineers handle source ingestion and cleaning; MLOps Engineers configure Git and DVC integration.

Week 2: Data Pipeline & Preprocessing Standardization

- **Tasks:** Finalize corpus preprocessing workflows, validate language-tag consistency across subsets, and establish automated data ingestion pipelines into DVC tracking.
- **Deliverables:** Standardized Preprocessing Pipeline, Version-Controlled Raw & Processed Datasets.
- **Roles:** Data Engineers write ingestion scripts; MLOps Engineers establish automated DVC data-pipeline stages.

Week 3: Experimentation Infrastructure & Evaluation Ground Truth

- **Tasks:** Implement and test various text chunking strategies. Build the synthetic query set to serve as the evaluation ground truth. Define MLflow tracking conventions and experiment logging schemas.
- **Deliverables:** Synthetic Query Evaluation Dataset (Ground Truth), Chunking Strategy Benchmark Suite, Configured MLflow Tracking Environment.
- **Roles:** ML Engineers develop chunking strategies and evaluation benchmark scripts; MLOps/Devs configure the MLflow server and tracking protocols.

Week 4: Embedding Model Benchmarking & RAG Harness

- **Tasks:** Execute core ML comparison across multilingual embedding models. Train and evaluate the optional intent classifier. Construct the initial working Retrieval-Augmented Generation (RAG) harness.
- **Deliverables:** Multilingual Embedding Model Evaluation Report, Intent Classifier Model Artifacts (Optional), First Functional RAG Baseline Harness.
- **Roles:** ML Engineers execute embedding benchmarks and build the intent classifier; Systems Engineers assemble the core RAG harness.

Phase 2 — Productionization & Deployment (Weeks 5–7)

Week 5: Model Registry & FastAPI Service Initialization

- **Tasks:** Register top-performing model artifacts into the MLflow Model Registry. Develop a lightweight FastAPI service wrapping the RAG pipeline to create a testable local backend interface.
- **Deliverables:** Registered Model Artifacts, Working FastAPI Service Endpoint (Local).
- **Roles:** MLOps Engineers manage registry staging; Backend Engineers build and expose FastAPI endpoints.

Week 6: Service Testing, Risk Auditing & Containerization

- **Tasks:** Execute service unit and integration tests. Perform system risk and safety checks in parallel. Package the FastAPI application and dependencies using Docker.
- **Deliverables:** System Integration & Risk Assessment Audit Report, Standardized Production Docker Image.
- **Roles:** QA/Security Engineers conduct risk checks; Backend/MLOps Engineers author Dockerfiles and verify container builds.

Week 7: CI/CD Pipeline & Live Deployment

- **Tasks:** Configure CI/CD pipelines to automate testing and container deployment. Deploy the live API endpoint to the hosting environment and execute automated smoke tests.
- **Deliverables:** Production CI/CD Pipeline Configuration, Live Deployed API Endpoint, Smoke Test Validation Suite.
- **Roles:** DevOps/MLOps Engineers manage CI/CD runner configurations and final endpoint deployment.

Phase 3 — Monitoring, Feedback & Governance (Weeks 8–10)

Week 8: Observability & Drift Infrastructure Setup

- **Tasks:** Instrument request/response logging across the active endpoint. Integrate Evidently AI to capture production telemetry and begin drafting the visual monitoring dashboard.
- **Deliverables:** Integrated Endpoint Logging System, Functional Evidently AI Data Collector, Preliminary Monitoring Dashboard UI.
- **Roles:** MLOps Engineers integrate logging middleware; Frontend/Data Engineers assemble monitoring dashboard components.

Week 9: Drift Simulation & Model Card Draft

- **Tasks:** Execute a deliberate data/concept drift experiment to test monitoring alerts. Draft the system Model Card using safety/risk findings from Phase 2 and per-language metrics from Phase 3.
- **Deliverables:** Controlled Drift Test Suite, Draft Model Card & Governance Spec.
- **Roles:** ML/MLOps Engineers run drift simulations; Governance/ML Leads draft the Model Card.

Week 10: Retraining Loop Closure & Final Delivery

- **Tasks:** Complete the full automated operational loop: detect drift → trigger retrain → evaluate/compare candidate→ redeploy live service. Finalize and publish the completed Model Card.
- **Deliverables:** Fully Validated Retraining Loop, Final System Model Card, End-to-End Production Infrastructure.
- **Roles:** MLOps Engineers execute the automated loop validation; All Leads finalize governance documentation and handoff.

Critical Path: Dataset & DVC Setup → Chunking & Ground Truth → Multilingual Model Benchmarking → Model Registry & FastAPI Harness → Live Deployment → Drift Simulation & Automated Retraining Loop.